## Health-Project 3: Breast Cancer Detection

# Experiment 1: Objective Definition Document

**Course:** CSE4001 — Applications of Machine Learning in Industry | **Target:** Malignant (1) vs. Benign (0)

## 1. Problem Statement & Clinical Context

A diagnostic pathology laboratory processes high volumes of Fine-Needle Aspirate (FNA) biopsy samples for suspected breast cancer. Manual microscopic examination of nuclear morphological features is labor-intensive and subject to inter-observer variability. The objective is to formulate an automated second-opinion machine learning triage system to analyze digitized morphological features of cell nuclei and automatically flag likely-malignant tumours for priority pathologist review.

## 2. Machine Learning Formulation

| Dimension | Specification |
|---|---|
| ML Experiment Type | Supervised Binary Classification |
| Input Features | 30 continuous nuclear morphological measurements (mean, SE, worst of 10 geometric features) |
| Target Variable | Diagnosis: 0 = Benign, 1 = Malignant (Binary Flag) |
| Primary Metric | **Malignant Class Recall / Sensitivity (≥ 98%)** to minimize missed malignancies |
| Secondary Metrics | Precision (≥ 90%), ROC-AUC (≥ 0.98), F1-Score, False Negative Rate (FNR) |
| Deployment Latency | < 200 ms per sample via RESTful FastAPI scoring service |

## 3. Clinical Success Metrics & Rationale

In clinical oncology screening, a **False Negative** (classifying a malignant tumor as benign) carries fatal risk, as vital cancer therapy is delayed. Conversely, a **False Positive** results in routine pathologist review. Therefore, the ML optimization objective strictly prioritizes Malignant-class Recall over raw overall accuracy.

## 4. Stakeholder Analysis

- **Pathologists & Cytotechnologists:** Clinical end-users receiving prioritized alerts and confidence scores.
- **Patients:** Benefit from accelerated turnaround times and reduced diagnostic error.
- **Laboratory Directors:** Ensure clinical QA, HIPAA/GDPR regulatory compliance, and auditability.
- **ML Engineering Team:** Responsible for training, validating, deploying, and monitoring the predictive pipeline.

## 5. Operational Constraints & Governance

- **Second-Opinion Decision Support:** The system acts as assistive decision support and does not replace certified pathologist review.
- **Class Imbalance Management:** The ~63% benign to 37% malignant ratio requires threshold tuning and loss weighting.
- **Zero Data Leakage:** Preprocessing statistics must strictly derive from training partitions.